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Effects of a Physical Training Programme on Cognitive Function and Walking Efficiency in Elderly Persons with Dementia

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Key Words
Alzheimer · Physical training · Cognitive function

Abstract
Objectives: To study the effects of physical stimulation based on walking exercises, equilibrium and endurance on cognitive function and walking efficiency in patients with dementia. ***Methods:*** Randomized controlled trial including 31 subjects suffering from dementia (age: 81.8 ± 5.3 years). The intervention group (n = 16) benefited from a 15-week physical activity programme involving three 1-hour sessions per week. The control group (n = 15) did not practice any physical activities. Before and after rehabilitation, all subjects were evaluated with the Rapid Evaluation of Cognitive Functions test (ERFC French version) and walking analysis. ***Results:*** After the 15 weeks of rehabilitation, the subjects from the intervention group improved their overall ERFC score ($p < 0.01$), while those in the control group decreased their overall ERFC score. Interactions were also observed between walking parameters and groups ($p < 0.01$); the intervention group improved walking capacities through heightened walking speed, stride length and a reduction in double limb support time. Lastly, the subjects from the control group presented a reduction in both walking speed and stride length. ***Conclusion:*** This study shows that a physical activity programme can slow cognitive decline and improve quality of walking in elderly persons suffering from dementia.

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Introduction

Alzheimer's disease constitutes a major phenomenon in modern society. Associated in its initial stages with memory disorders, Alzheimer's is a neurodegenerative disease that brings about a variety of cognitive disorders and motor perturbations [1, 2].

Alzheimer-type dementia presents degenerative cognitive deficits. Subsequent to the loss of memory, the deficits carry over into the areas of language (aphasia), motion organization (apraxia), visual recognition (agnosia) and the executive functions (such as making decisions and planning ahead). Also, life style may modulate the effects of pathological aging on cognitive functions. As one of the factors related to life style, physical training is presently the subject of an appreciable body of research. The effects of physical activity on healthy aged people have been amply demonstrated, especially regarding cognitive functioning [3]. Moreover, other studies have

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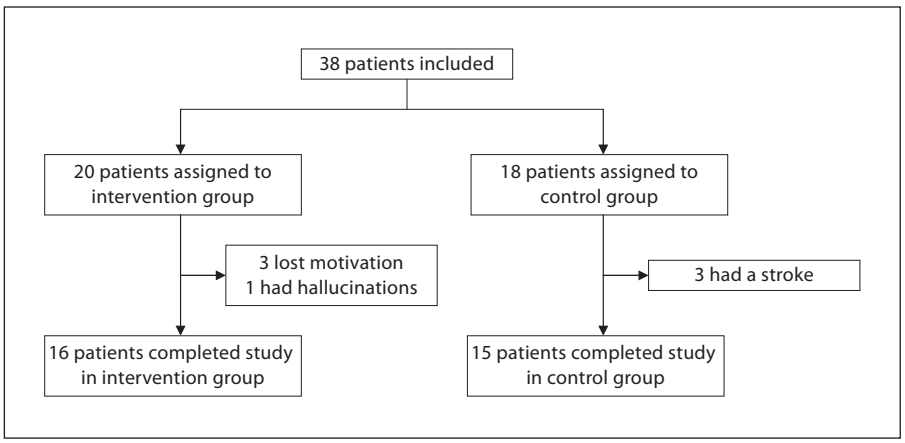

Fig. 1. Flowchart of enrolled patients.

shown that a physically active life style provides protection against Alzheimer's [4, 5]. It consequently appears that leisure-time activities, social networks and physical activities have a positive impact on these people's cognition in the overall framework of disease prevention. For elderly people with dementia, on the other hand, studies are few and far between. In 2004, one meta analysis [6] had synthesized published results on the effects of exercise training for cognitively impaired older adults. Exercise training was associated with positive effects on health, physical, cognitive, functional and behavioral components. It would seem that physical activity slows down the effects of pathological aging by maintaining and even enhancing the executive functions, which can be defined as a variety of high-level cognitive processes allowing for behavior that is flexible and context-adapted [7], communication abilities [8] and the cognitive functions [9, 10].

Even though Alzheimer-type dementia entails the progressive loss of the cognitive functions, motor system disorders are part and parcel of the clinical picture. In comparison with the healthy elderly, recordings of the way Alzheimer people walk have highlighted various modifications shortened stride, diminished speed, and heightened average limb support time [1, 2, 11]. The loss of locomotor efficiency in dementia patients increases the risk of falls [12–14]. As a result, improvement of the spatiotemporal parameters of walking appears primordial. Rolland et al. [15] have shown that a physical activity programme speeded walking.

The primary aim of this study was to influence of physical training on the cognitive functions of elderly people suffering from dementia. The second aim was to demonstrate that this training can also influence locomotor efficiency.

Methods

Participants

Inclusion of the participants was based on two criteria: one was cognitive, a diagnosis of Alzheimer-type dementia by a neurologist according to DSM IV criteria with a score on the Mini Mental State Examination (MMSE) [16] lower than 23; the other was motor-based skills, i.e. the ability to walk 10 m without technical assistance. A total of 38 people over 75 years of age from the same nursing home (81.8 ± 5.3 years) were included in this study (fig. 1). Subjects were randomized into two groups using a permutation table, subsequent to the consent, if necessary, of their legal guardian. The two groups were comprised of a 'physical activity' intervention group (IG), consisting of 20 persons, and 18 persons in a control group (CG).

The 20 participants in the IG were enrolled in a physical activity programme. All subjects (IG and CG) had the same participation in manual and intellectual activities organized by the nursing home (pottery, painting, soft gymnastics, outings, etc.).

Four patients withdrew from the IG group after a few months. Three stopped due to lost motivation, and one stopped due to a psychotic disorder (hallucinations). Three CG patients missed the post-test evaluation because of a health complication (stroke). Among the 38 patients initially present, only 31 completed the whole series of initial and final evaluations.

Walking Assessment

The analysis of walking was performed with a Bessou locometer [17] and SATEL® software. The principle of this data analysis method consists of recording the longitudinal movement of each foot during an approximately 10 m long walk. The movement of each foot was transmitted, by means of a thread, to an optical sensor. The spatiotemporal data of a walking cycle were recorded. The phases of acceleration and deceleration were eliminated to ensure that the calculated values corresponded to the stabilized speed [17]. Prior to walking, the patient was seated on a chair to facilitate strap installation. The patient's feet were in contact with the locometer, and the threads were switched on. Following these verifications, the patient stood up and walked the 10 m at a comfortable speed. A chair was set up at that dis-

Table 1. Results of ERFC score and walking parameters before and after 15 weeks of rehabilitation in the control group and the intervention group

	Before rehabilitation		After 15 weeks of rehabilitation	
	control group	intervention group	control group	intervention group
Cognitive function				
ERFC	28.33 ± 7.11	26.81 ± 6.42	23.23 ± 8.37	30.38 ± 7.66[a]
Gait parameters				
Walking speed, m/s	0.91 ± 0.20	0.74 ± 0.12	0.75 ± 0.16	1.02 ± 0.24[a]
Stride length, m	1.00 ± 0.20	0.93 ± 0.20	0.86 ± 0.15	1.02 ± 0.22[a]
Double limb support time, s	0.13 ± 0.04	0.16 ± 0.04	0.14 ± 0.04*	0.11 ± 0.03*, [a]

Data are expressed as means ± SD.
[a] Changes are significantly different ($p < 0.01$) from those observed in the control group.
* Significantly different from data observed before training ($p < 0.01$).

tance to provide a demented individual with a reference point and to facilitate removal of the straps. The parameters selected were walking speed, stride length and double limb support time.

Cognitive Functions Assessment

The subjects benefited from an overall analysis of the cognitive functions with the French ERFC (Rapid Evaluation of Cognitive Function) [18]. The test consisted of 12 sub-tests: spatial orientation, attention span, immediate and deferred memory, mental calculation, reasoning and judgment, comprehension, denomination, repetition, a written order, verbal fluency, apraxia, visual decoding and writing. The maximum score is 50, and a score lower than 46 indicates a significant probability of cognitive deficit. In this way, the test allowed us to appreciate the cognitive functions taken as a whole.

Physical Training

Our study covered a period of 19 weeks, the first 2 and the last 2 being reserved for the cognitive and motor-oriented assessments. During 15 weeks, the IG patients went through three 1-hour sessions per week. During the first 2 weeks, the participants were prepared for physical activity through exercises involving articular mobilization and muscle stimulation. Evaluation of individual aptitudes constituted the initial step of orienting content programming by defining an individual motor profile. Next, during 13 weeks, we elaborated a programme based upon exercises in walking, equilibrium and stamina. Three themes were taken up in each session in accordance with our objectives, and they occasioned 40 min of exercises per session. Each week, one session was given over to walking and the amelioration of walking parameters through motor route exercises (e.g. walking by striding over boards, going up a step, zigzagging). After that, a second weekly session was devoted to stamina exercises, particularly on an ergocycle with the arms and the legs. Participants could thereby keep up a light to moderate effort over a relatively long time span. During this effort, the patients were equipped with a Beurer P-100® cardiofrequency meter glove. While unable to carry out an effort test, we nonetheless managed to determine the targeted cardiac frequencies encompassed between 60 and 70% of reserve cardiac frequency.

Finally, the last session of the week involved activities (e.g. dance and stepping) that combined stamina, equilibrium and walking. This session was supposed to be enjoyable, as its objective was to maintain the motivation of one and all throughout the duration of the programme. The hour was divided into three parts: the first part consisting of 10 min of contact, articular mobilization and warm-up, the second part was 40 min of the previously described activities and then 10 min of return to calm and relaxation.

The follow-up necessitated preliminary programming of each session and allowed for activity content to evolve in accordance with each person's progress.

Statistics

The Kolmogorov-Smirnov test (K-S test) was used to test distribution normality for each parameter. The initial subject characteristics in the two groups were compared using the Student's t test for quantitative variables and using a McNemar χ^2 with a Yates correcting for qualitative variables.

Each participant was assessed before and after the rehabilitation, and comparisons between were made with a repeated measures ANOVA. A Spearman correlation test was used to study the relationship between the changes in cognitive score (Δ values) and the changes in walking variables (Δ values).

These analyses were made with Statistica 8, and the significance threshold was set at $p < 0.01$.

Results

Each variable was normally distributed (K-S test). The two groups (IG vs. CG) were similar regarding age (IG: 82.0 ± 5.8 years, CG: 81.7 ± 5.1 years), gender (IG: 12 women/4 men, CG: 11 women/4 men), MMSE (IG: 12.6 [7–20], CG: 12.9 [8–19]) and gait parameters (table 1).

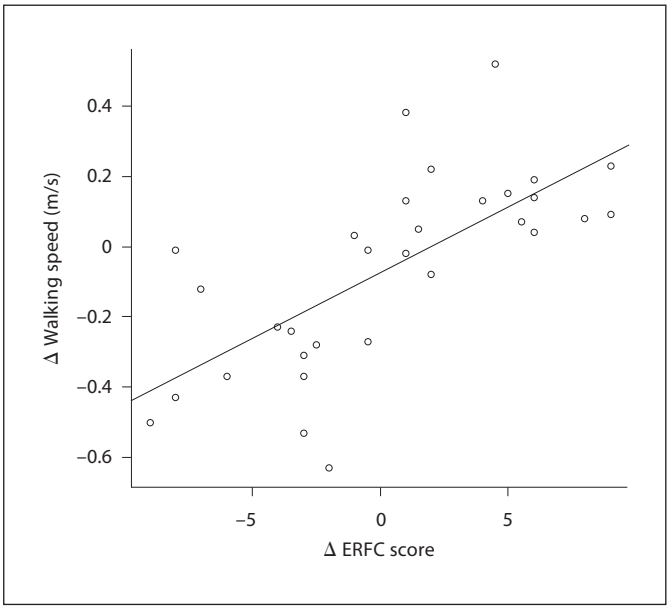

Fig. 2. Correlation between the changes in cognitive score (ERFC score) and walking speed in dementia patients. This correlation was found to be significant ($p < 0.001$).

Participation in the physical activity programme came to an average of 40.6 ± 9.0 of 45 sessions.

Analysis of Cognitive Performance

The results revealed significant interaction between evolution of the overall ERFC score and group ($F_{(1, 29)} = 55.1$, $p < 0.01$) (table 1).

Analysis of the Walking Parameters

Significant interactions were observed when analyzing the walking variables indicating that the walking speed ($F_{(1, 29)} = 53.4$, $p < 0.01$) and the stride length ($F_{(1, 29)} = 16.3$, $p < 0.01$) diminished in the CG and increased in the IG. Moreover, we observed that double limb support time significantly increased in the CG whereas it significantly decreased in the IG ($F_{(1, 29)} = 27.0$, $p < 0.01$).

Correlation between Cognitive Score and Walking Parameters

We observed a significant correlation between changes in the ERFC score and walking speed ($\rho = 0.76$, $p < 0.01$) (fig. 2), stride length ($\rho = 0.61$, $p < 0.01$) and double limb support time ($\rho = -0.74$, $p < 0.01$).

Discussion

This study shows that a 15-week physical activity programme, based on walking, equilibrium and stamina exercises, induced a positive influence on the spatiotemporal variables concerning walking and cognitive capacities in aged persons suffering from dementia. Moreover, the changes of those two kinds of variables (gait and walking variables vs. cognitive score) were significantly correlated.

These improvements can reasonably be attributed to the added amount of physical exercise. However, they may in part stem from an increased amount of personal contact (or variation in nursing home routine or some expectation of global improvement). The number of participants was also quite low. Medication and co-morbidities were unknown, and this may have affected the locomotion abilities. Randomization was included in the design of our study and should have buffered such an effect. However, given the relatively small sample size, it is certainly possible that these variables were not equally distributed between groups (17% of the control group and 0% of the intervention group suffered a stroke).

Finally, it was complicated to really standardize the amount of activity in these patients.

However, as regards the cognitive functions, our results are congruent with those of the literature [9, 10, 15, 19]. Lindenmuth and Moose [19] were the first to demonstrate the effect of somatic exercises and relaxation on 43 patients suffering from Alzheimer-type dementia. Somewhat later, two studies showed that the stamina exercises (walking or ergocycle) improved the overall cognitive functions evaluated by the MMSE pertaining to persons suffering from Alzheimer-type senile dementia [9, 15]. Lautenschlager et al. [10] highlighted a slight amelioration of the cognitive functions assessed by means of the ADAS-cog. Following the literature, stamina-based activities (e.g. walking and ergocycle) were presented as practices essential for improving the cognitive functions in healthy aged people [3] and in demented aged people as well [9, 15], particularly through the improvement of aerobic capacity.

The hypothetical mechanisms underlying the relation between physical activity and cognitive functioning in aged persons and those suffering from dementia are neither well known nor formally established. Several hypotheses have been put forward. The first one involves the nutritive influence of physical activity on the synaptic and/or neuronal function [20–22]. The second hypothe-

sis is based upon a correlation between aerobic capacity and cognitive performances [23].

Alzheimer's is generally considered as a disease mainly affecting cognitive functions and sparing motor function. Van Doorn et al. [12] drew attention to the fact that Alzheimer patients are twice as likely to fall as healthy aged persons. The falls generally take place while the subject is walking, which is to say over the course of a dynamic activity. As for our study, it shows a positive impact of such activity on locomotor efficiency in our patients, which may consequently diminish the risk of falling, as was shown by Toulotte et al. [24] in the healthy elderly.

Links between physical exercises and walking and equilibrium capacities in Alzheimer patients have already been observed. According to Rolland et al. [25], significantly heightened walking speed has been observed after 6 months of walking for 12 months with supervision. The results of Santana-Sosa et al. [26] confirmed the hypothesis that structured physical activity over a 12-week programme significantly improves muscle tone, flexibility, agility, dynamic equilibrium, stamina and walking itself as well as lowers the risk of falls. Even though there exists little literature in this field, a number of authors have spotlighted the interest of a physical exercise programme with regard to improved walking parameters in elderly persons not suffering from cognitive decline [24, 27, 28]. Walking parameters in Alzheimer patients present some predictive indicators of falls (shortened stride length, slowing down, increased average double limb support time) [1, 2, 11]. Therefore, it appears that finding ways to improve those variables in these patients is of high importance.

Even though the elements used to analyze static equilibrium do not apply to dynamic equilibrium, one may nevertheless suppose that improved static equilibrium helps to optimize walking. Lord et al. [29] showed that in the falling elderly, physical activity based on equilibrium and muscle reinforcement improves equilibrium with eyes both open and closed, thereby reducing the number of falls. Toulotte et al. [24] mentioned the positive effect of a physical exercise programme on unipodal tests, once again eyes both opened and closed. Control loops serve to maintain rhythm and ensure the adaptation of the different walking parameters to environmental constraints. These loops are of proprioceptive, vestibular and visual origin. Improved equilibrium should consequently allow for optimized walking.

As a complement to this hypothesis, improved walking may stem from increased muscle tone. In the nursing home, a large number of patients are 'deconditioned', and their unfit state favors a sedentary way of life in which the muscles are underused.

Conclusion

Alzheimer's disease represents a serious health problem. Moreover, today's main medical treatments are essentially symptomatic and remain highly controversial. In this context, our study has shown that a physical activity programme is doubly beneficial, both slowing cognitive decline and improving locomotor efficiency. To conclude, organization of a professionally supervised and regularly accessible physical activity workshop should be included in the treatment and care of Alzheimer patients in nursing homes. However, further research in this area is warranted as no intensity, frequency, duration or type of activity has been validated for optimized improvement.

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